



• 100% FREE · Live Cohort

👋 Hey beginners — this one's for you

Get started with Cloud & AI on AWS in just 5 days.

A friendly, **beginner-friendly intro** to AWS — no prior experience needed. Perfect for students, graduates, and anyone curious about cloud or AI.

- ✓ **Learn the essentials** — cloud, storage, compute, AI services
- ✓ **Hands-on demo on Day 5** — see AWS services working together
- ✓ **10 hours total** · live · 3 timezones to choose from

★★★★★

"Concepts are explained in a simple, step-by-step manner. Even advanced AWS topics feel clear and manageable."

Gururaj Rathod · DevOps Engineer

★★★★★

"I still go back to it as a reference. So much content — theory plus hands-on with real-world examples."

Antonio Villarán López · Cloud Architect

LIVE DATES

May 25 – 29

Mon to Fri · 2026 · 2 hrs/day

PICK YOUR TIMEZONE

EST 9:00 – 11:00 AM

GMT 2:00 – 4:00 PM

IST 6:30 – 8:30 PM

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FOR BEGINNERS · NO EXPERIENCE NEEDED

CLOUD & AI ON AWS



DAY 3 · Compute: **EC2** · **Lambda** · **ECS**

MAY 25-29 · 2 HRS/DAY · **LIVE**

Agenda

Day 1 ● What is cloud computing, AWS and Identity & access (IAM)



Day 2 ● Introduction to Storage (Amazon S3), and content delivery (AWS CloudFront)



Day 3 ● Compute (EC2, Lambda, ECS) and how services communicate (SQS, SNS)

Day 4 ● A friendly intro to AWS AI services: Bedrock, Rekognition, Polly, Textract

Day 5 ● Hands-on demo: Lambda, Textract, Bedrock, S3 and CloudFront chained together



Introduction to Amazon EC2

Virtual Machines in the cloud.

What is Amazon EC2?

Think of it as renting a computer in AWS's datacentre — you get the machine; you control what runs on it.



EC2 in plain English

"A virtual computer running in AWS's data centre. You choose the OS, the CPU, the memory, and the storage. You connect to it over the internet and run whatever you like — a web server, a database, a script."

- Runs in an AWS Availability Zone
- You pay by the second (or hour)
- Start, stop, resize any time
- Full root / admin access to the OS



The laptop analogy

"Imagine buying a laptop, but instead of it sitting on your desk, it lives in AWS's building. You remote in over the internet and use it just like your own machine — install software, run code, host a site."

The key difference from your laptop:

- It never needs charging or rebooting
- You can upgrade it in minutes
- You can clone it instantly
- You only pay when it's switched on

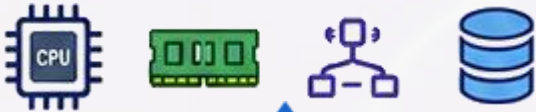
WHAT IS A VIRTUAL MACHINE & VIRTUALIZATION?

Virtualization lets one physical server run multiple isolated virtual machines (VMs), each with its own operating system and resources.

Virtual Machine 1



Windows OS

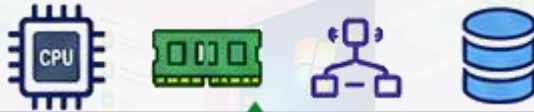


Virtual Machine 2

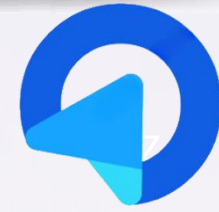
Linux



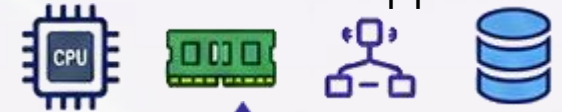
Linux OS



Virtual Machine N



Other OS / App

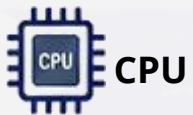


Hypervisor



The hypervisor sits between the VMs and the physical server. It allocates and manages CPU, memory, network, and storage for each VM, keeping them isolated from each other.

Physical Server (hardware)



CPU

(Shared by VMs)



Memory

(Shared by VMs)



Network

(Shared by VMs)



Storage

(Shared by VMs)



EC2 Instance Types

Every instance belongs to a family — pick the one that matches your workload.

 General Purpose t3, t4g, m6i <i>The all-rounder. Good balance of CPU, memory and network.</i> USE Websites, small databases, dev environments t3.micro (free tier eligible)	 Compute Optimised c6i, c7g <i>Heavy CPU work. Great when your app crunches a lot of data fast.</i> USE Gaming servers, batch processing, ML inference c6i.large	 Memory Optimised r6i, x2 <i>Huge amounts of RAM. For workloads that need to hold data in memory.</i> USE In-memory databases, real-time analytics, SAP r6i.large	 Storage Optimised i3, d3 <i>Fast local disk throughput. For apps that read/write enormous datasets.</i> USE Data warehouses, Hadoop, distributed file systems i3.large	 Accelerated Computing p4, g5, inf2 <i>GPU-powered. For machine learning training, video rendering, HPC.</i> USE Deep learning, 3D rendering, scientific simulation g5.xlarge
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Amazon Machine Images (AMIs)

An AMI is the blueprint your EC2 instance is built from.



What is an AMI?

"Think of an AMI like a USB stick with a pre-installed operating system and software. You plug it into a new EC2 instance and it boots up exactly as configured — instantly ready."

An AMI contains:

- Operating system (Linux, Windows, etc.)
- Pre-installed software & configurations
- EBS snapshot (the root volume)
- Launch permissions and block device mapping

AWS Managed AMIs

Official images from AWS — Amazon Linux 2023, Ubuntu, Windows Server. Always up to date and free to use.

AWS Marketplace AMIs

Pre-configured images from vendors — WordPress, Nginx, SQL Server. Some are free, others have a licence cost.

Your Own Custom AMIs

Build your own. Launch an instance, configure it, then save a snapshot as an AMI. Clone it as many times as you like.

When would you actually use EC2?

EC2 gives you full control — use it when you need the flexibility, not just as a default.



Use EC2 when...

- ✓ **Your app needs a specific OS** — custom Linux distros, Windows Server
- ✓ **Long-running workloads** — a process that runs 24/7 for weeks
- ✓ **Lift & shift migrations** — moving an on-premise server to cloud as-is
- ✓ **You need root access** — install custom software, kernel modules
- ✓ **Legacy apps** — software that only runs on a specific OS version
- ✓ **Stateful applications** — apps that need persistent local disk (games, databases)



Consider alternatives when...

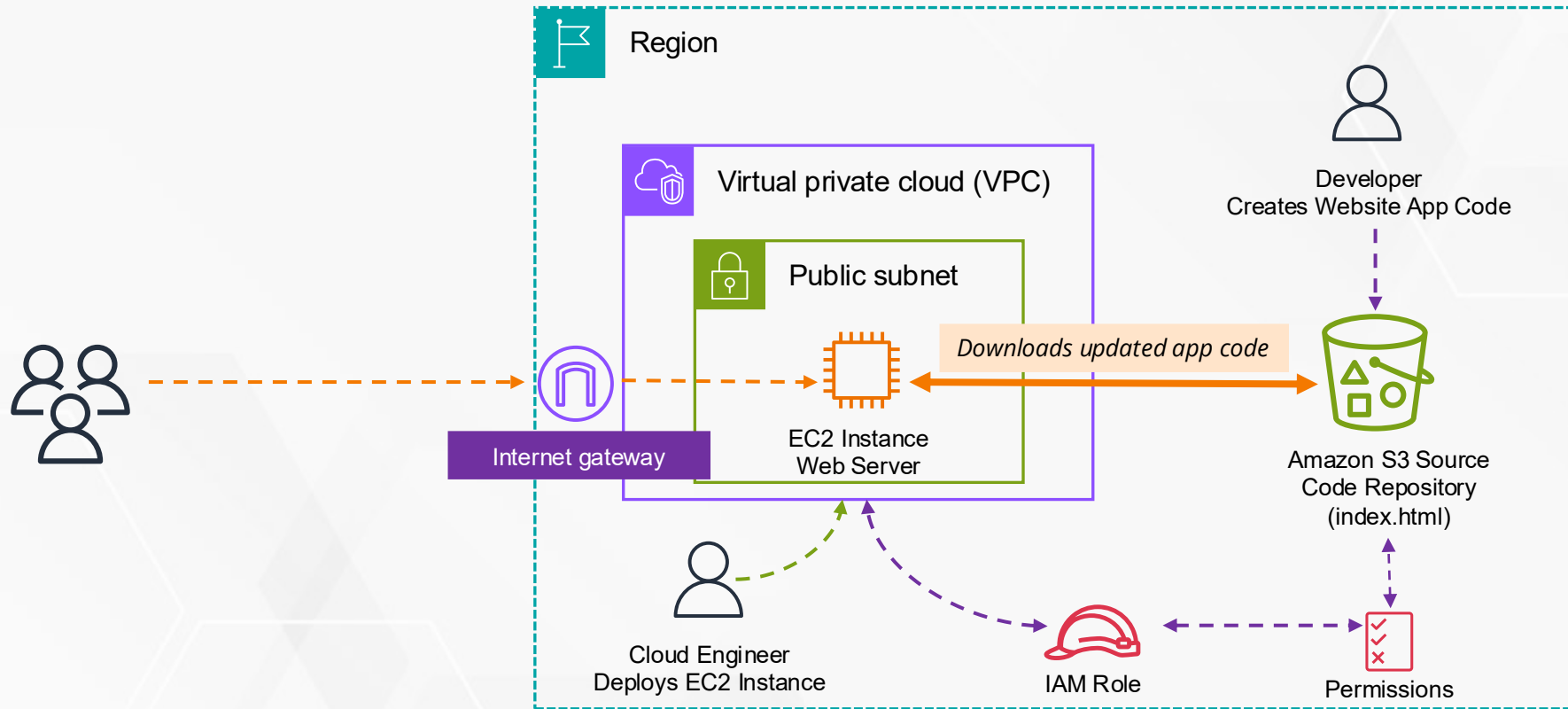
- ✗ **Short-lived tasks** → Lambda (serverless — runs for seconds)
- ✗ **Containerised apps** → ECS or EKS (managed containers)
- ✗ **Fully managed databases** → RDS (no OS to manage)
- ✗ **Static websites** → S3 + CloudFront (no server needed)
- ✗ **Event-driven processing** → Lambda + SQS (scales to zero)
- ✗ **Serverless APIs** → Lambda + API Gateway



Launch an EC2 Instance

Host a resume website

A simple architecture



Automation Script

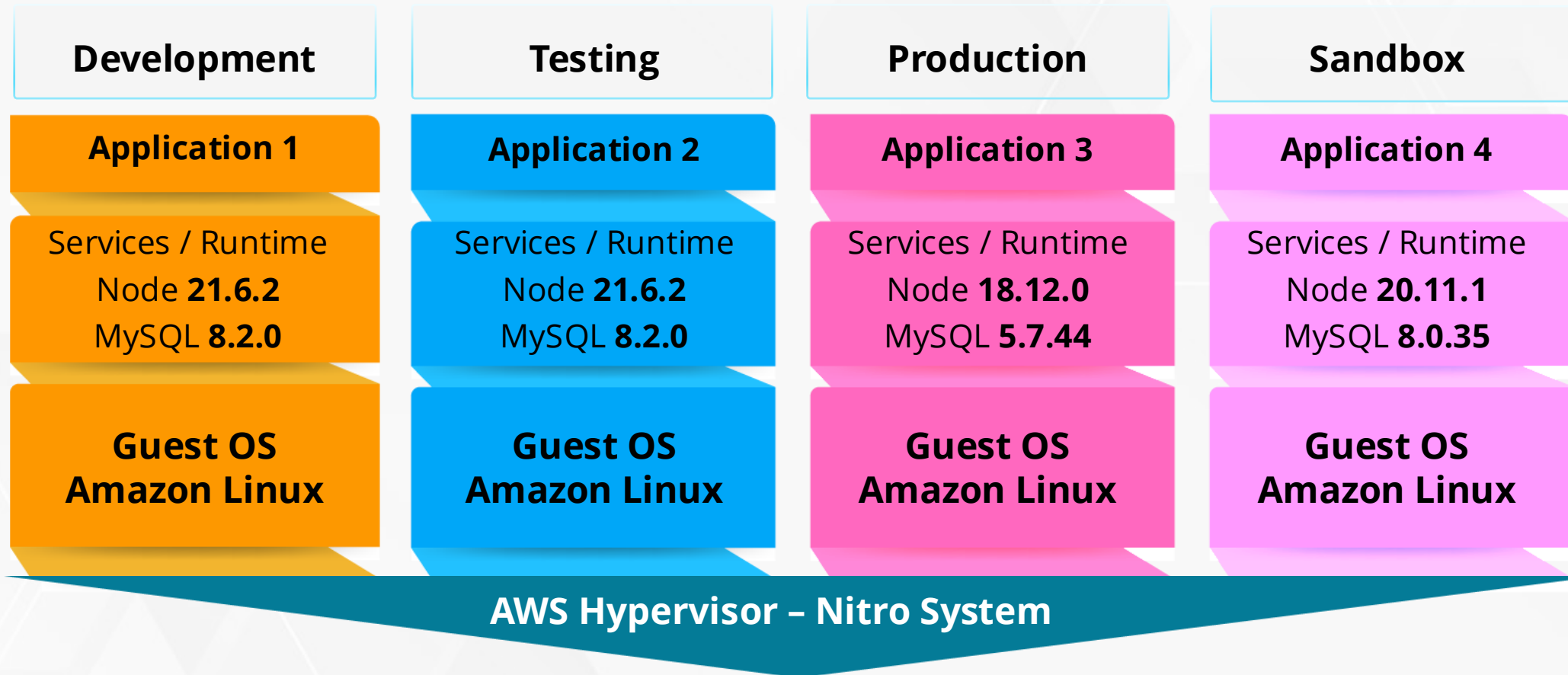
- Installation of webserver software
- Installation of any databases
- Automate the deployment of app code to webserver



Amazon ECS

Containers on AWS

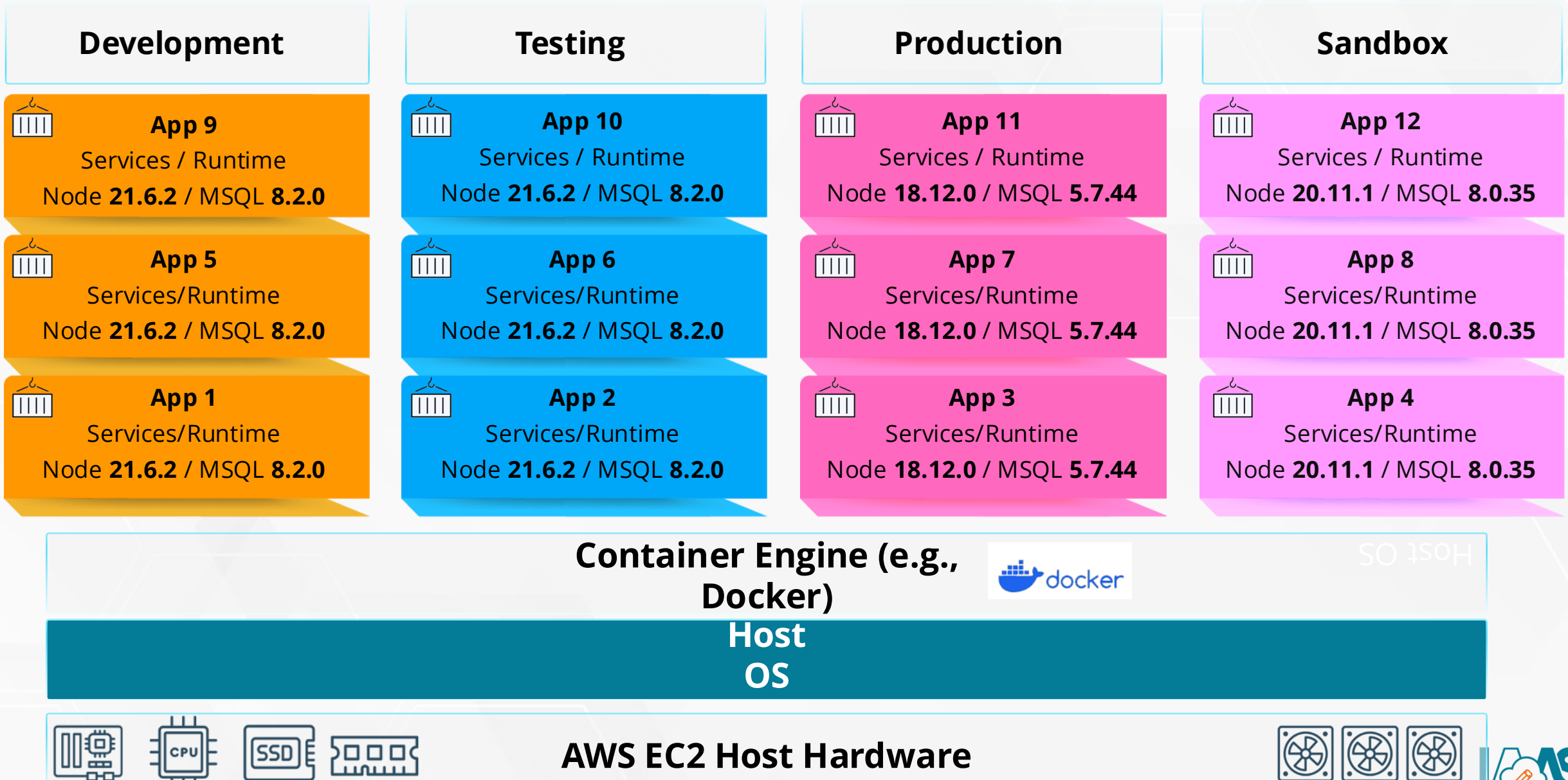
Quick Recap on Virtualization



AWS EC2 Host Hardware



Application Virtualization - Containerization

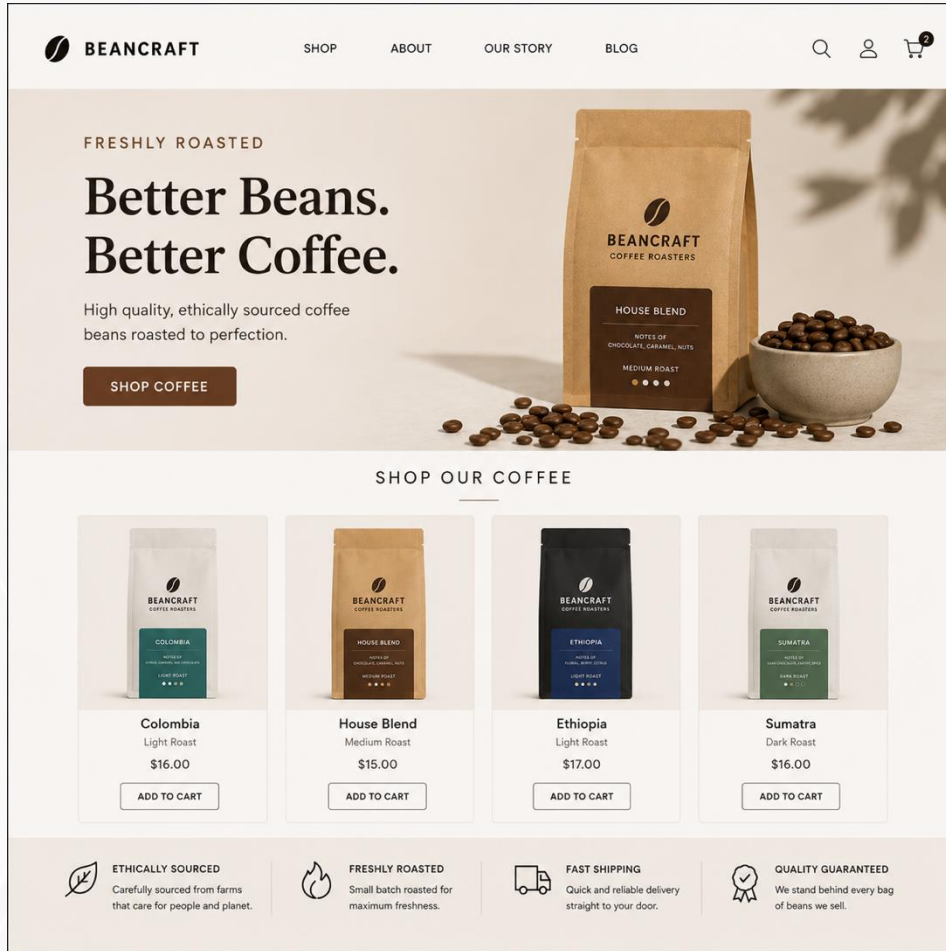




AWS Lambda

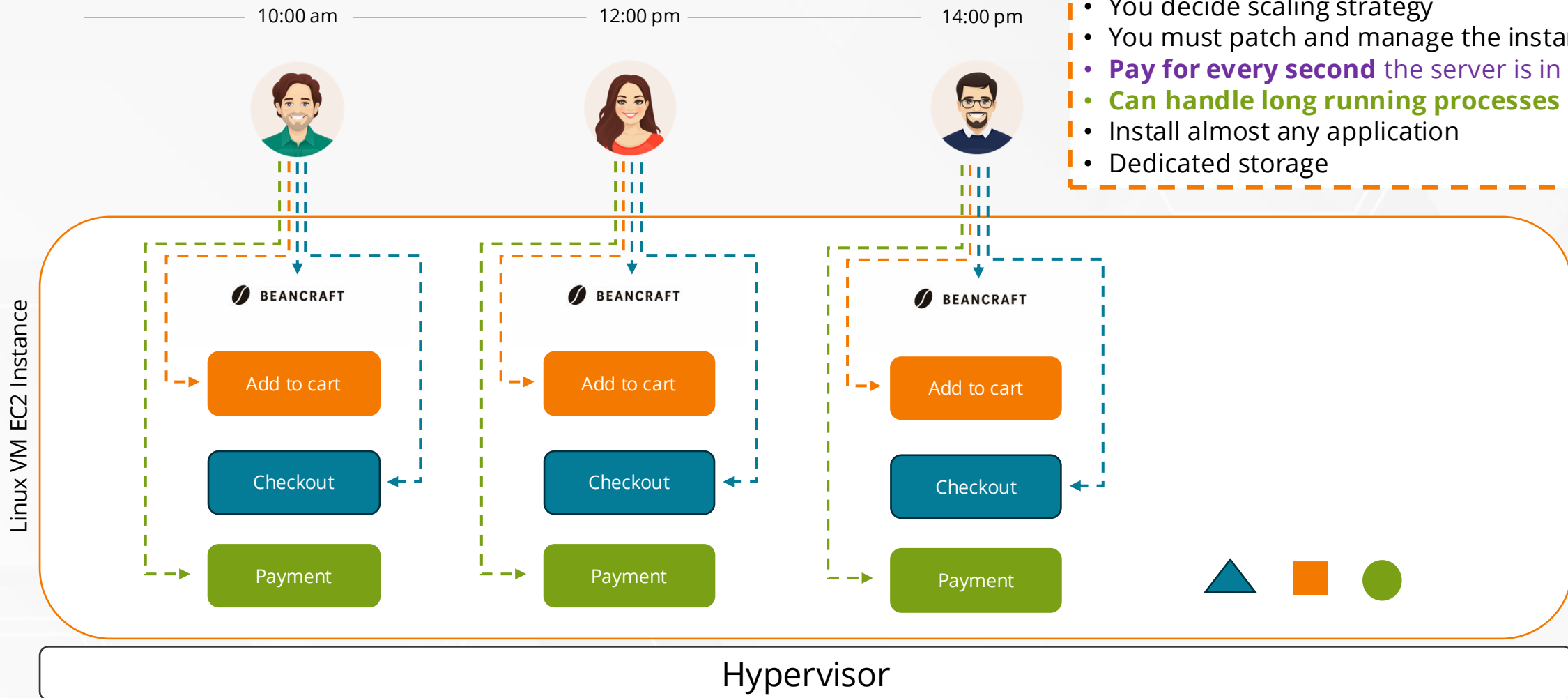
Serverless Event-Driven Compute

The Scenario



- Coffee Shop Business
- Ecommerce website
- Customers place orders at different times of the day – random
- Website offers standard ecommerce service
 - Browse products
 - Add to cart
 - Checkout
 - Payment gateway

Traditional EC2 deployment



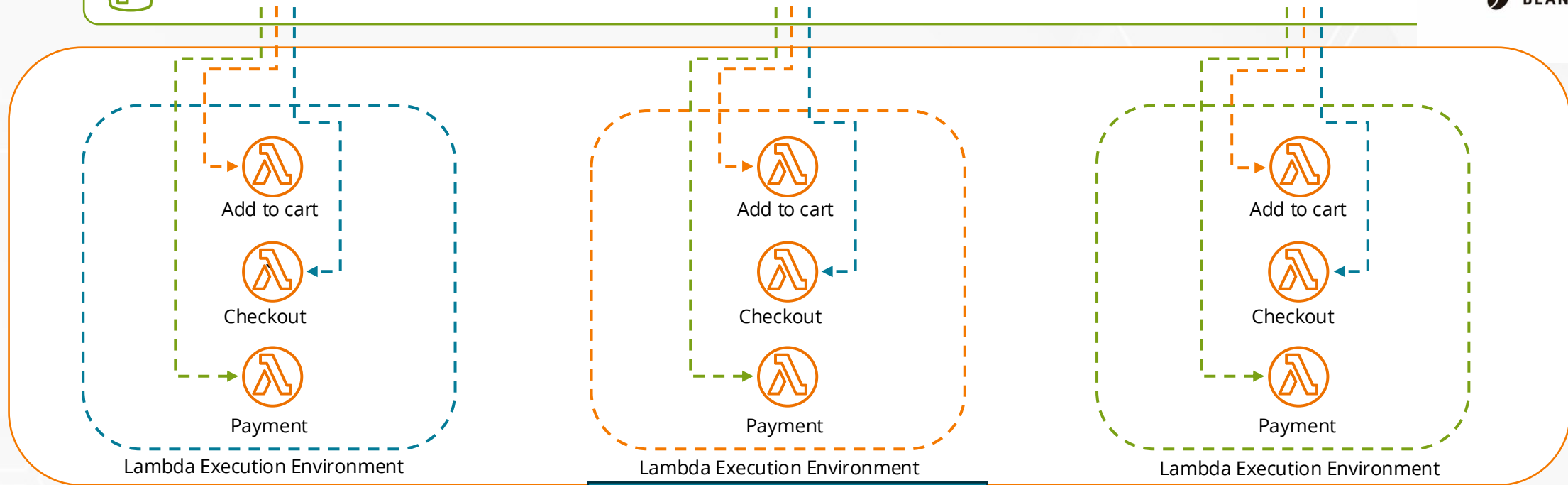
- Dedicated Instance running until you terminate
- **Choice of OS** (Linux or Windows)
- You decide scaling strategy
- You must patch and manage the instance
- **Pay for every second** the server is in the 'On' state
- **Can handle long running processes**
- Install almost any application
- Dedicated storage

What is AWS Lambda?

- Serverless execution environment managed entirely by AWS
- No choice of underlying OS (choose runtime only: Java, python etc.,
- AWS automatically scales based on incoming requests
- AWS handles patching, maintenance, and infrastructure management
- Pay only for requests and execution time (milliseconds), not idle time
- Best suited for short-lived event-driven workloads (max 15 minutes per invocation)
- No dedicated persistent local storage



BEANCRAFT



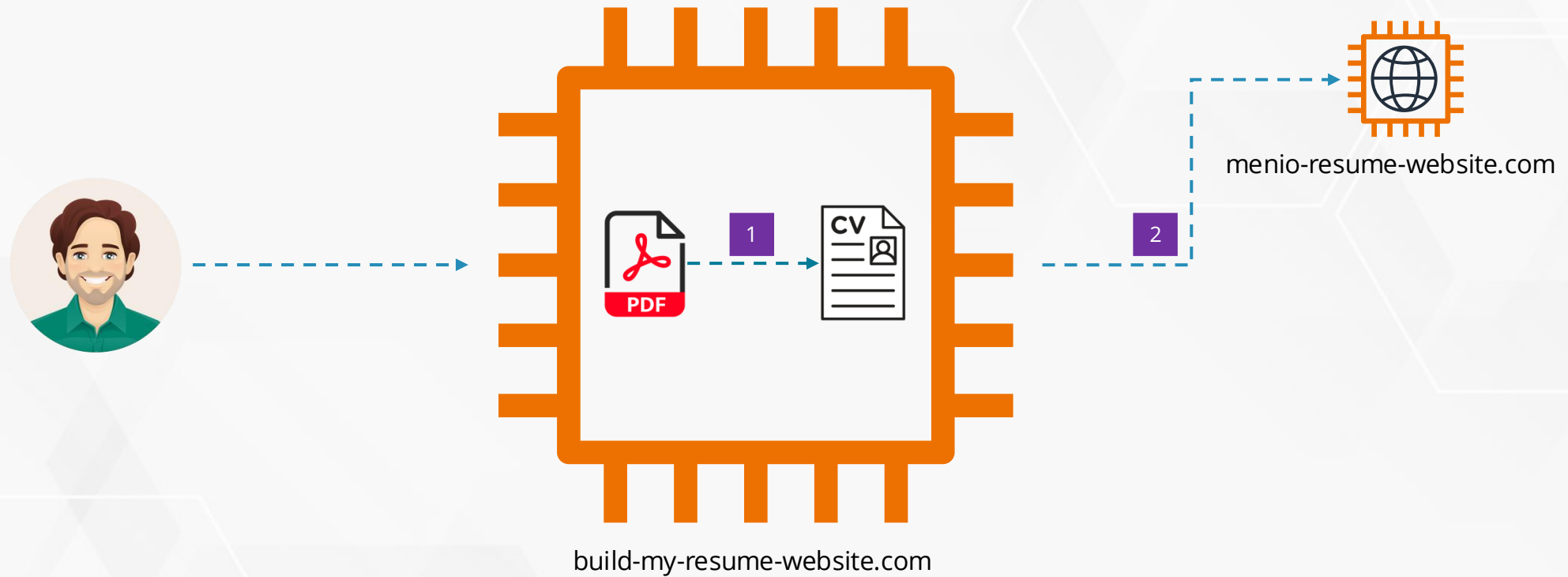
microVMs (AWS Managed in Sandbox)

Firecracker (virtual machine monitor (VMM))

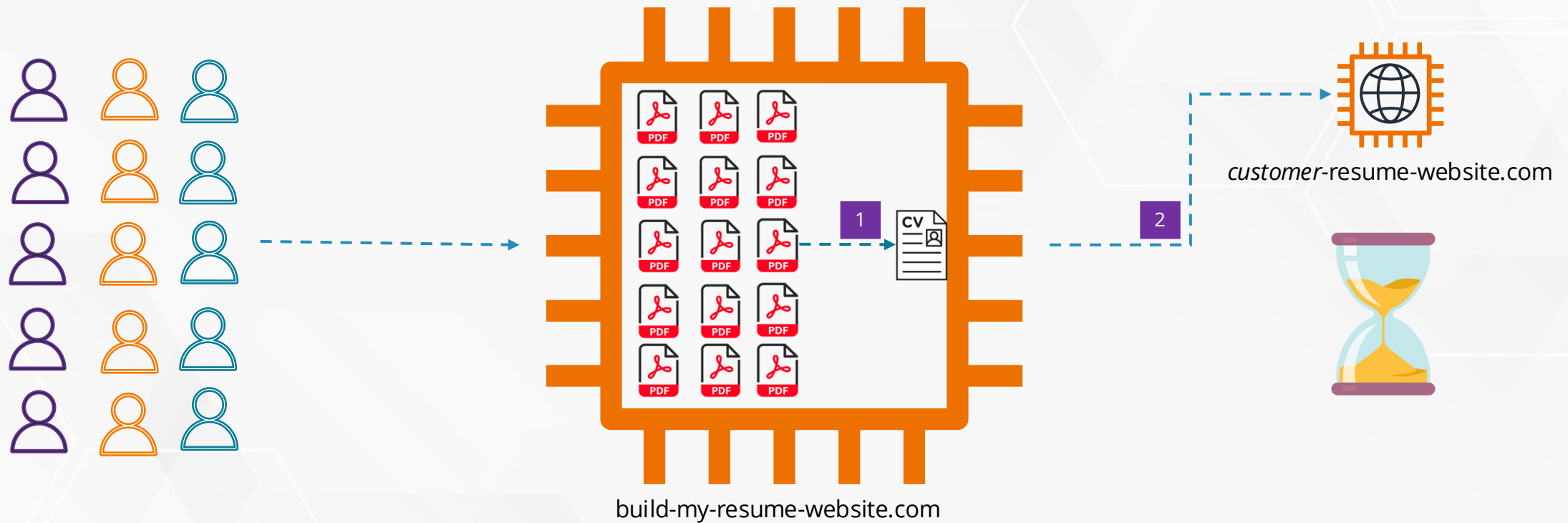


Decouple Architecture

The need to decouple frontend from backend

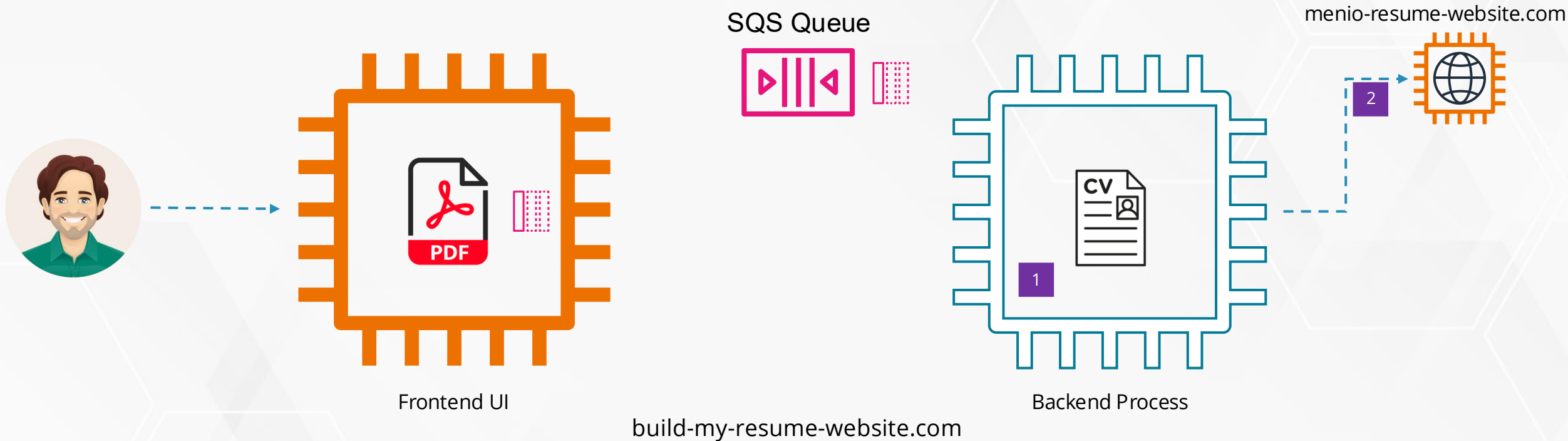


The need to decouple frontend from backend



The need to decouple frontend from backend

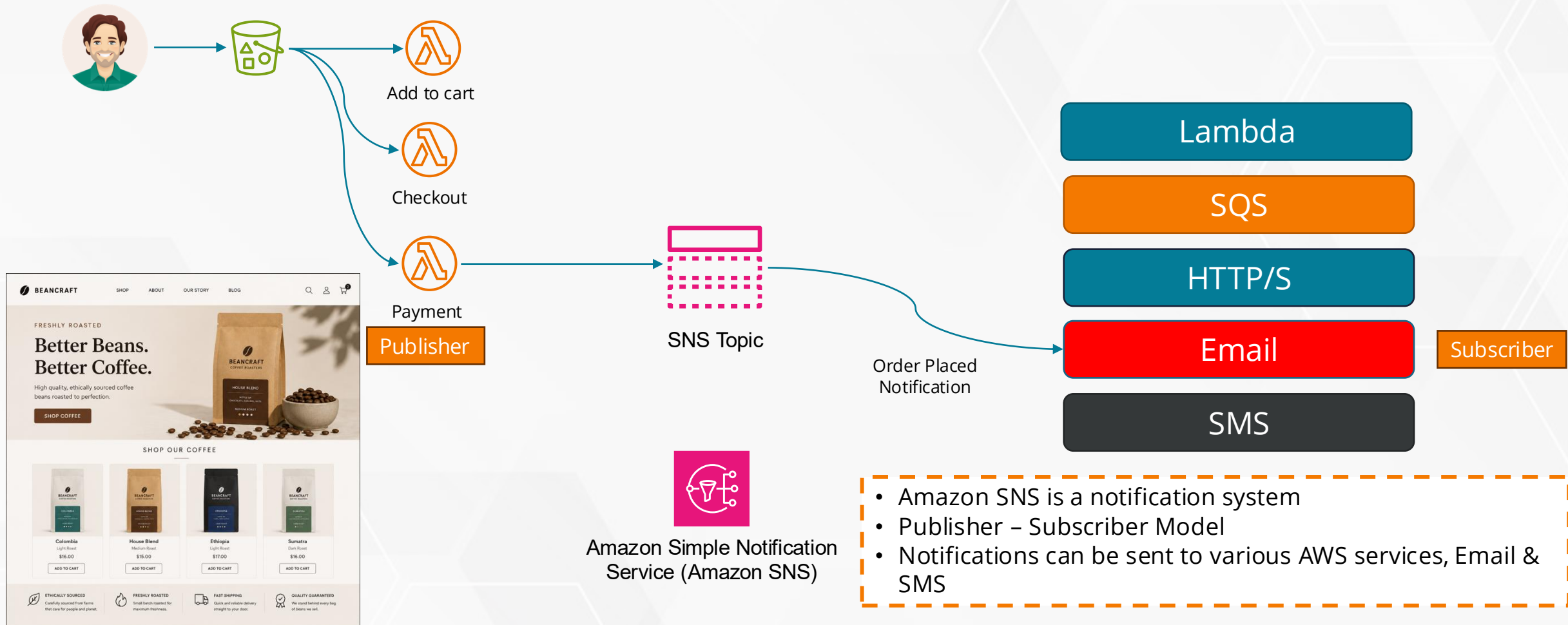
Introducing Amazon Simple Queue Service (Amazon SQS)



- Amazon SQS is a message queuing system
- Messages are stored in the queue (default 4 days)
- **Backend retrieves messages when capacity available**
- Backend processes request
- Frontend and Backend scale independently

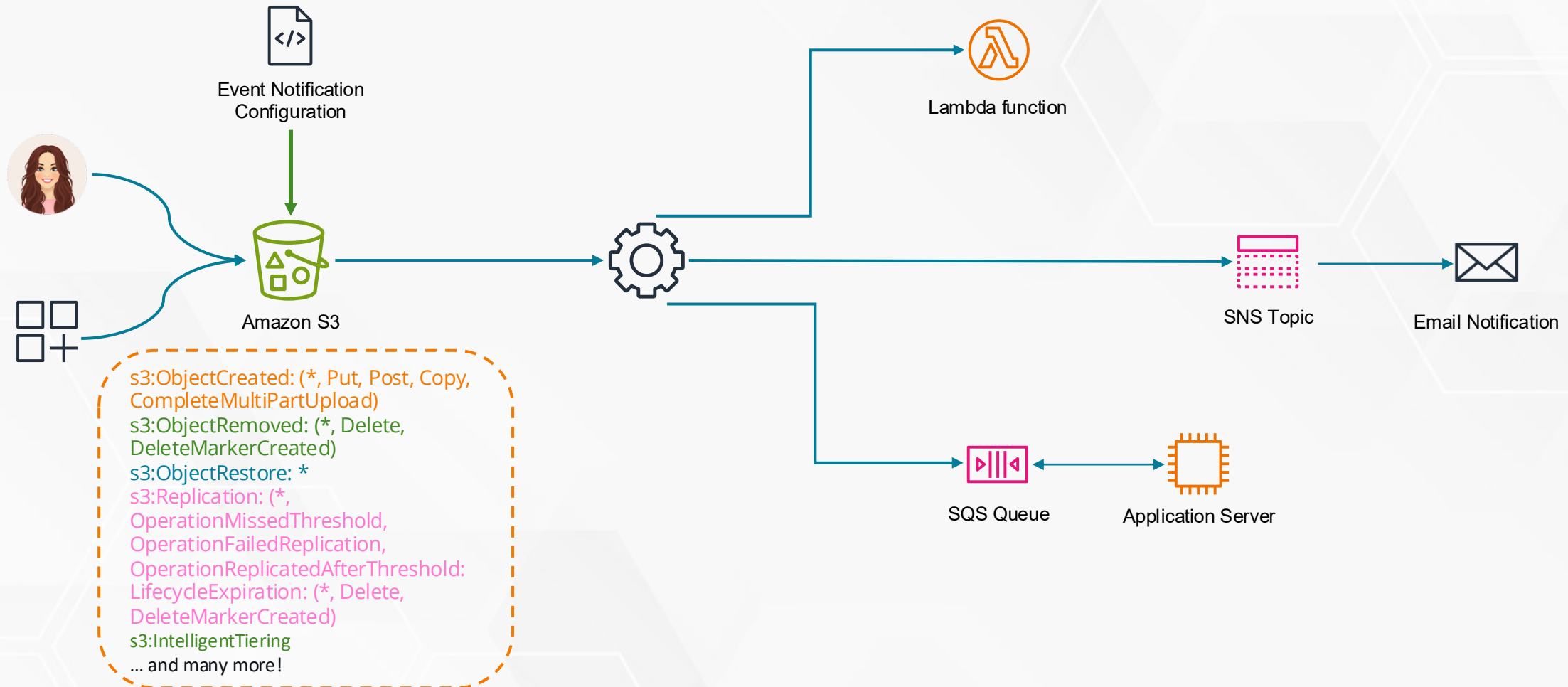
The need to decouple frontend from backend

Introducing Amazon Simple Notification Service (Amazon SNS)



Amazon S3 Event Notification

Use the Amazon S3 Event Notifications feature to receive notifications when certain events happen in your S3 bucket





Homework

- Deploy an EC2 Instance
- Deploy a web server and publish the sample resume website

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